Tao Huang

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ACADEMIC APPOINTMENTS

Assistant Professor in Water Resources Engineering, Aug. 2025 - Present
Department of Civil and Environmental Engineering, Idaho State University, Pocatello, ID

Postdoctoral Research Associate in Earth Systems Data Science, Jul. 2023 - Aug. 2025
Earth Systems Science Division, Pacific Northwest National Laboratory, Richland, WA

EDUCATION

Ph.D. in Civil Engineering, Purdue University (2019 - 2023)
M.S. in Hydraulic Engineering, Tsinghua University (2012 - 2014)
B.S. in Hydraulic and Hydropower Engineering, Tsinghua University (2008 - 2012)

RESEARCH INTERESTS

- Physics-Based and Data-Driven Modeling of Hydrologic Extremes
- Uncertainty Quantification and Probabilistic Flood and Drought Modeling
- Hydropower Systems and Resilient Water Infrastructure
- Artificial Intelligence and Data Science for Water Resources Management

RESEARCH GRANTS

High-Resolution Probabilistic Flood Mapping for High-Hazard Aging Dams in Southeastern Idaho
Idaho Water Resources Research Institute | **PI** | $54,835 | Sep. 2026 - Sep. 2027

Impacts of Hydrometeorological Data Gaps on Drought Characterization
Idaho Water Resources Research Institute | **Co-PI** | $68,000 | Sep. 2026 - Sep. 2027

Developing Probabilistic Flood Maps for Idaho Under Precipitation Uncertainty Using a Computationally Efficient 2D Hydrodynamic Model
Idaho Water Resources Research Institute | **PI** | $46,784 | Nov. 2025 - Nov. 2026

PUBLICATIONS & RESEARCH PRODUCTS

Peer-Reviewed Journal Articles

Huang, T., Salalila, A., Mueller, R., Martinez, J., Titzler, S., Lu, J., Renholds, J., Anderson, K., Trumbo, B., & Deng, D. (2026). Balancing Power Generation with Ecological Impact: Hydraulic and Biological Performance of a Modernized Adjustable-Blade Kaplan Turbine Assessed Using Sensor Fish Technology. *Energy Nexus*, 100878. Doi: https://doi.org/10.1016/j.nexus.2026.100878

Huang, T., Zang, X., Kondyukov, G., Hou, Z., Peng, G., Pander, J., Knott, J., Geist, J., Melesse, M. B., Jacobson, P. & Deng, D. (2025). Towards automated and real-time multi-object detection of anguilliform fishes from sonar data using YOLOv8 deep learning algorithm. *Ecological Informatics*, 103381. Doi: 10.1016/j.ecoinf.2025.103381

Huang, T., Salalila, A., Meyers, A., Fu, T., Martinez, J., Hou, H., & Deng, D. (2025). Velocity- and pressure-based metrics for estimating strike injuries during fish passage through hydro turbines, *Results in Engineering*, 104535. Doi: 10.1016/j.rineng.2025.104535

Huang, T., & Merwade, V. (2024). Reply to comment on "improving Bayesian model averaging for ensemble flood modeling using multiple Markov chains Monte Carlo sampling" by Jasper Vrugt. *Water Resources Research*, 60(11), e2024WR037387. Doi: 10.1029/2024WR037387

Huang, T., & Merwade, V. (2024). Developing customized NRCS unit hydrographs for ungauged watersheds in Indiana, *Journal of Hydrologic Engineering*, 29(4), 04024022. Doi: 10.1061/JHYEFF.HEENG-6089

Huang, T., & Merwade, V. (2024). Beyond a fixed number: Investigating uncertainty in popular evaluation metrics of ensemble flood modeling using bootstrapping analysis, *Journal of Flood Risk Management*, 17(2), e12982. Doi: 10.1111/jfr3.12982

Huang, T., & Merwade, V. (2023). Improving Bayesian model averaging for ensemble flood modeling using multiple Markov Chains Monte Carlo sampling. *Water Resources Research*, 59(10), e2023WR034947. Doi: 10.1029/2023WR034947

Huang, T., & Merwade, V. (2023). Uncertainty analysis and quantification in flood insurance rate maps using Bayesian model averaging and hierarchical BMA. *Journal of Hydrologic Engineering*, 28(2), 04022038. Doi: 10.1061/JHYEFF/HEENG-5851 (**Editor's Choice** and **Best Technical Paper**)

Huang, T., & Zhan, C. (2019). 2D FEM calculation of pre-stressed pier with an anchor block reserved slot. *Journal of North China University of Water Resources and Electric Power (Natural Science Edition)*, 40(2): 84-89+96. Doi: 10.19760/j.ncwu.zk.2019027

Huang, T., & Cheng, P. (2018). Structure calculation of overflow sluice piers. *Water Resources Planning and Design*, 2018(12): 192-198. Doi: 10.3969/j.issn.1672-2469.2018.12.050

Zuo, Z., Hu, Y., Li, Q., Li, B., & **Huang, T**. (2015). Temperature monitoring during concrete setting through cooling pipe monitors. *Journal of Tsinghua University (Sci & Technol)*, 55(1): 21-26. Doi: 10.16511/j.cnki.qhdxxb.2015.01.004

Du, B., Du, G., **Huang, T.**, Liu, M., & Yue, Y. (2014). Temperature and stress analysis of concrete face plate in construction period. *Journal of China Three Gorges University (Natural Sciences)*, 36(3): 23-27. Doi: 10.13393/j.cnki.issn.1672-948X.2014.03.006

Huang, T., Chen, Q., Hu, Y., Zhong, Q., & Liu, T. (2013). Research of XYPEX on the property of concrete crack repair. *Concrete*, 2013 (11): 88-92. Doi: 10.3969/j.issn.1002-3550.2013.11.024

Manuscripts Under Review

Munna, M., **Huang, T.**, Savage, B., Daniel, B., & Li, X. (2026). Probabilistic Flood Inundation Mapping Under Rainfall Uncertainty Using a Computationally Efficient 2D Hydrodynamic Model. *Journal of Hydrology*.

Bhusal, S., & **Huang, T**. (2026). Convergent Evidence of Long-Term Hydrologic Change Across Idaho from Streamflow, Water Availability, and Drought Indicators. *Journal of the American Water Resources Association*.

Technical Reports

Hou, H., **Huang, T**., Salalila, A. L., & Deng, Z. (2025). Hydropower Biological Evaluation Toolset (HBET) Version 3.0: User Guide. Technical Report. Doi:10.2172/3000007

Hou, H., **Huang, T**., Salalila, A. L., & Deng, Z. (2025). HBET V3.0 Installation Manual. Technical Report. Doi:10.2172/2999986

Huang, T., & Merwade, V. (2023). Developing customized NRCS unit hydrographs (Finley UHs) for ungauged watersheds in Indiana (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2023/10). West Lafayette, IN: Purdue University. Doi: 10.5703/1288284317644

TEACHING & MENTORING EXPERIENCE

Idaho State University | Assistant Professor | 2025 - Present

CE4499/5599 Data Analysis Methods in Water Resources; CE2210 Engineering Statics — Fall 2026
CE3351 Engineering Hydrology — Spring 2026
CE4499/5599 Data Analysis Methods in Water Resources — Fall 2025

Purdue University | Graduate Teaching Fellow / Teaching Assistant | 2019 - 2023

CE340 Hydraulics — Spring 2021, Fall 2021 & Spring 2023
CE343 Elementary Hydraulics Laboratory — Fall 2020

Tsinghua University | Student Counselor| 2012 - 2013

Acted as a Student Counselor for 90 students, overseeing the Undergraduate Freshman Training Program.

HONORS & RECOGNITION

Co-developer of HBET V3.0 (2025), Pacific Northwest National Laboratory; HBET was part of the **2026 R&D 100 Award-winning Sensor Fish Suite + HBET technology**.

Best Technical Paper Award (Feb. 2024), Environmental & Water Resources Institute (EWRI).

Editor's Choice (Feb. 2023), Journal of Hydrologic Engineering

Lyles Teaching Fellowship (Jan. 2023), Purdue University.

Jacques Delleur Travel Award (Dec. 2022), Purdue University.
Dorothy Faye Dunn Fellowship (Oct. 2022), Purdue University.
C4E Travel Grant (Apr. 2022), Purdue University.
Yeh Travel Fellowship (Nov. 2021), Purdue University.
Zhang Guangdou Scholarship (Jul. 2014), Tsinghua University.
Undergraduate Mentor Certificate of Honor (Jul. 2013), Tsinghua University.
Outstanding Undergraduate Thesis (Jun. 2012), Tsinghua University.

PROFESSIONAL SERVICE

Grant Proposal Reviewer: National Science Foundation (NSF).
Journal Peer Reviewer: Water Resources Research, Journal of Hydrology, Journal of Hydrologic Engineering, Journal of Flood Risk Management, Scientific Reports, Natural Hazards and Earth System Sciences, Applied Sciences, Computers.
Conference Session Convener: AGU Fall Meeting 2026, H11C: Advancing Flood Characterization, Modeling, and Communication.

CONFERENCE PRESENTATIONS

Huang, T., Daniel, B., & Li, X. (Dec. 2025). Probabilistic Flood Mapping for Idaho Under Precipitation Uncertainty Using a Computationally Efficient 2D Hydrodynamic Model. Poster presentation at American Geophysical Union (AGU) Fall Meeting in New Orleans, LA.
Huang, T., Hou, H., Salalila, A., Martinez, J., & Deng, D. (Aug. 2025). Estimating Strike Injuries During Fish Passage Using Enhanced Hydropower Biological Evaluation Toolset. Oral presentation at American Fisheries Society (AFS) Annual Meeting in San Antonio, TX.
Huang, T., Martinez, J., Fu, T., Salalila, A., & Deng, D. (Sep. 2024). Metrics for Estimating Strike Injuries During Fish Passage through Kaplan Turbines. Oral presentation at AFS Annual Meeting in Honolulu, HI.
Huang, T. (Dec. 2023). Investigating streamflow variability of HUC-2 regions in the Conterminous United States from 2003 to 2022 water years. Poster presentation at AGU Fall Meeting in San Francisco, CA.
Huang, T., & Merwade, V. (Dec. 2022). An integrated evaluation framework based on generalized likelihood uncertainty estimation for quantifying uncertainty in flood modeling. Oral presentation at AGU Fall Meeting in Chicago, IL.
Huang, T., & Merwade, V. (Dec. 2022). Estimating Bayesian model averaging weights and variances of ensemble flood modeling using multiple Markov chains Monte Carlo. Poster presentation at AGU Fall Meeting in Chicago, IL.
Huang, T., & Merwade, V. (Sep. 2022). Developing customized NRCS unit hydrographs for ungauged watersheds in Indiana. Oral presentation at the Indiana Association for Floodplain and Stormwater Management Annual Conference in South Bend, IN.
Huang, T., & Merwade, V. (May 2022). Evaluating uncertainty in FEMA flood insurance rate maps using Bayesian model averaging (BMA) and hierarchical BMA. Oral presentation at American Water Resources Association (AWRA) Geospatial Water Technology Conference in Austin, TX.
Huang, T., & Merwade, V. (May 2022). Developing customized NRCS unit hydrographs for ungauged watersheds in Indiana, USA. Oral presentation at AWRA Geospatial Water Technology Conference in Austin, TX.
Huang, T., & Merwade, V. (Dec. 2021). Evaluating uncertainty in modeling of FEMA flood insurance rate maps using Bayesian model averaging (BMA) and hierarchical BMA. Oral presentation at AGU Fall Meeting in New Orleans, LA.
Huang, T., & Merwade, V. (Dec. 2021). Developing customized NRCS unit hydrographs for ungauged watersheds in Indiana, USA. Poster presentation at AGU Fall Meeting in New Orleans, LA.

INDUSTRY EXPERIENCE

Civil Engineer | PowerChina Kunming Engineering Corporation Limited

Kunming, China | Jul. 2014 - Jun. 2019

- **Thalpitigala Reservoir Project** | Sri Lanka

Chief Design Engineer; led a six-member team in 3D dam design and stakeholder coordination.

- **Integrated Landslide Mitigation Project** | Sri Lanka

Chief Design Engineer; led a five-member team in landslide mitigation design and slope stability analysis.

- **Nam Ou-4 & Nam Mang I & Nam Khan 3 Hydropower Projects** | Laos

Designed dam spillways and piers; served as an on-site engineering representative for one year.

- **Corumana Dam Project |** Mozambique

Conducted structural analysis and design of prestressed dam piers and collaborated with AECOM engineers.

TECHNICAL SKILLS

Programming: Python, R, MATLAB, Visual Basic, Julia

Database: SQL (Azure SQL Server, Oracle SQL Developer)

Tools: High-Performance Computing, Azure, TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy

Research Techniques: Bayesian Analysis, Uncertainty Quantification, Machine Learning, Deep Learning

PROFESSIONAL AFFILIATIONS

American Geophysical Union | 2021-Present

American Society of Civil Engineers | 2022-Present

American Society for Engineering Education | 2026-Present